

WASP-10b

R-0811

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_181030 · created 2026-07-28T12:01:07+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458421.696 +/- 0.0042 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.208 +/- 0.032
Transit depth (Rp/Rs)^2	4.31 +/- 1.33 [%]
Orbital Inclination (inc)	89.84 +/- 1.3
Airmass coefficient 1 (a1)	1.012 +/- 0.043
Airmass coefficient 2 (a2)	-0.01 +/- 0.1
Scatter in the residuals of the lightcurve fit is	4.35 %
Best Comparison Star	#2 - [334, 144]
Optimal Aperture	2.33
Optimal Annulus	7.39
Transit Duration (day)	0.0969 +/- 0.0044

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.208 ± 0.032 vs 0.1592 (deviation 1.43σ)
Duration fit vs published	0.0969 d vs 0.0946 d (deviation 0.49σ)
Mid-time O–C	-9.6 min
Depth SNR	6.1
Residual scatter	4.35 %

Light curve

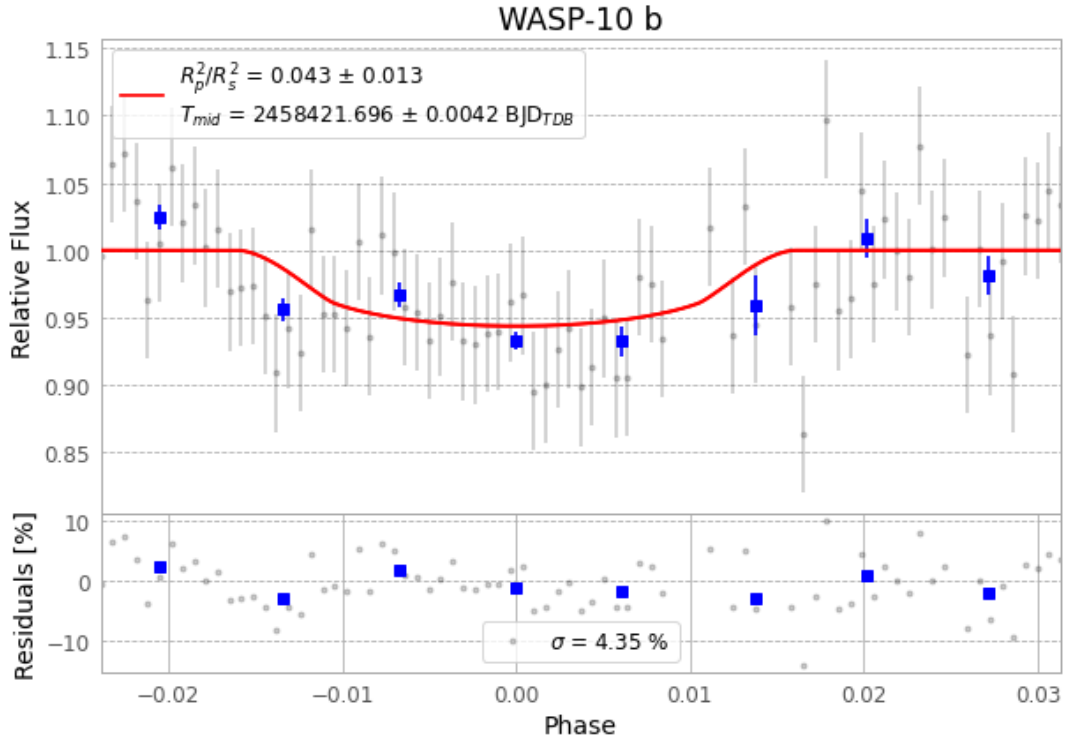


Figure 1 — FinalLightCurve_WASP-10 b_2018-10-29.png (SHA-256 d848d48f349f4087...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [338, 243], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 ba7f28a906a33123...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

83 FITS frames (55 MB), WASP-10181030024812.FITS → WASP-10181030065412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-181030.log (f506b3ae682c...), FinalLightCurve_WASP-10 b_2018-10-29.png (d848d48f349f...), FinalLightCurve_WASP-10 b_2018-10-29.csv (e29d2bc1dc77...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 12:01Z.